EXISTENCE OF THE SCHEME THEORETIC IMAGE OF A MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement begins with a slogan block (part of the counted statement span); the proof appeals to the largest quasi-coherent subsheaf statement for the general case.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The closed subspace defined by a quasi-coherent sheaf of ideals. Lemma lemma-closed-immersion-ideals, used at proof lines 3213-3215. Produces the closed subscheme in the quasi-coherent case.
- Existence of the largest quasi-coherent subsheaf of a submodule. Lemma lemma-largest-quasi-coherent-subsheaf (2809-2824), used at proof lines 3216-3219. Supplies the quasi-coherent ideal inside the kernel.
- The universal property of the scheme theoretic image. used at proof lines 3206-3208. Is what the constructed closed subspace is shown to satisfy.

Source reference. Original lemma 03MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4743–4743. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04CK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5694–5694. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. There exists a closed subspace $Z \subset Y$ such that f factors through Z and such that for any other closed subspace $Z' \subset Y$ such that f factors through Z' we have $Z \subset Z'$.*

Proof. Let $\mathcal{I} = \mathrm{Ker}(\mathcal{O}_Y \to f_*\mathcal{O}_X)$. If $\mathcal{I}$ is quasi-coherent then we just take Z to be the closed subscheme determined by $\mathcal{I}$, see Lemma 03MB. In general the lemma requires us to show that there exists a largest quasi-coherent sheaf of ideals $\mathcal{I}'$ contained in $\mathcal{I}$. This follows from Lemma 04CK. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.